

Securing Real-Time Mobile Analytics

Identity Binding and Low-Latency Authorization for Distributed Enterprise Data

ENTERPRISE SECURITY ARCHITECTURE

MOBILE-FIRST ANALYTICS

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The views expressed in this presentation are my own. They have not been reviewed or approved by Tesla



The Expanding Attack Surface



Enterprise and Operational leaders increasingly access enterprise analytics from **unmanaged personal devices**, not corporate laptops.

This shift expands the attack surface *before* a single dashboard even renders. The security architecture must account for this reality from the ground up.



BYOD access to operational data is now the norm, not the exception - and existing perimeter controls were not designed for it.



One Design Problem,
Not Three

Identity.
Authorization.
Performance.

Treating these as separate concerns creates dangerous gaps. This session presents a unified architecture where all three are resolved together.

Session Roadmap

01

Enterprise Identity Binding

Enforcing corporate trust on unmanaged personal devices

02

Role-Based Authorization at Speed

Cached permission checks balanced against strict BYOD access control

03

Cross-Domain Data Integration

Reconciling various departments, and workforce data securely

04

Data Freshness and Permission Consistency

Minute-level refresh without stale authorization state

05

AI-Augmented Insight Delivery

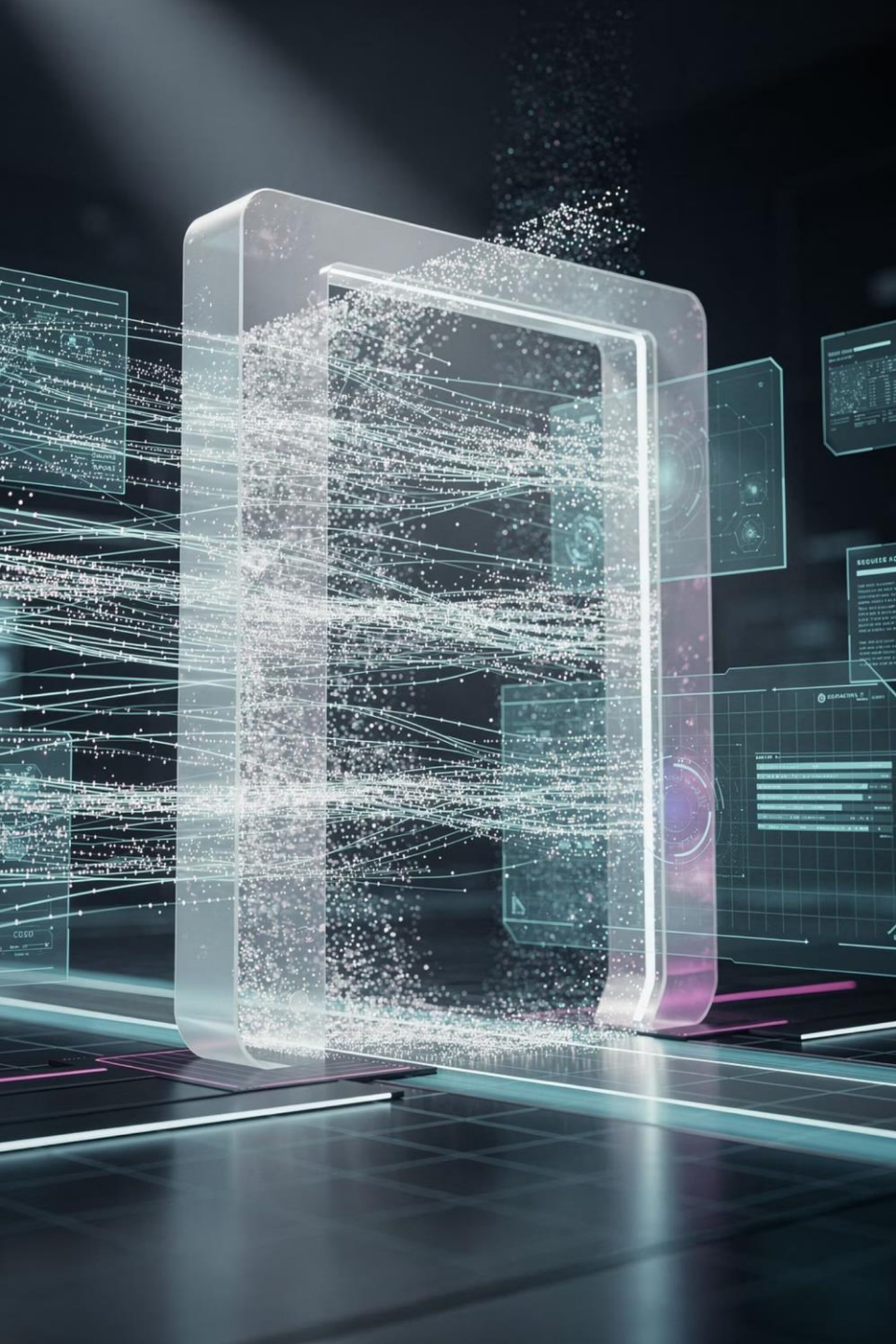
Query-scoped authorization for conversational and RAG-based access

Enterprise Identity Binding on Unmanaged Devices



A personal phone accessing operational data must carry the **same trust guarantees** as a managed corporate endpoint - achieved through layered identity binding rather than device management.

- **Device attestation:** OS integrity and hardware-backed key verification
- **Corporate identity federation:** SSO with certificate-backed token issuance
- **Continuous session validation:** Risk-signal re-evaluation on every request



Authorization Architecture

Role-Based Checks at Mobile Speed

On mobile, a slow authorization check is felt far more acutely than on desktop. The architecture must resolve permissions without adding perceived latency.

Balancing Strictness and Speed

Cached Permission Checks

Role resolution pre-computed and cached at the edge, keyed to identity token and data scope - eliminating round-trips on every render cycle.

TTL-Bounded Cache Invalidation

Strict time-to-live windows enforce re-validation when role assignments or resource policies change, preventing stale permissions from persisting.

Fail-Closed Fallback

When cache misses occur or network conditions are degraded, the system defaults to deny - never to permissive open access.

The BYOD context demands this strictness; the mobile UX demands the speed. Cached resolution delivers both without compromise.

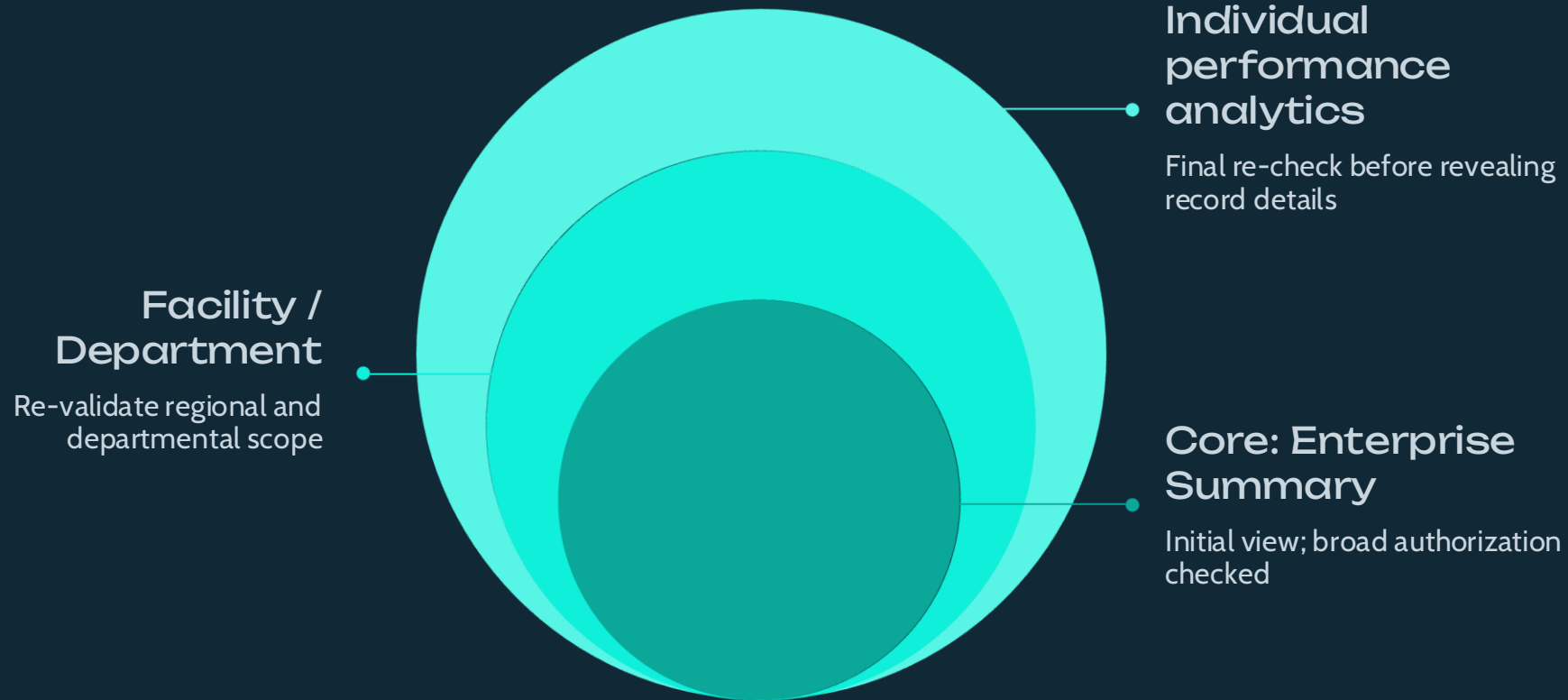
Cross-Domain Data Integration

A single mobile dashboard may surface data from **various department systems**- each governed by its own source system and access policy.

Reconciling these into a unified experience without creating permission gaps requires explicit policy federation at the integration layer, not the presentation layer.

 Policy must be enforced where data is joined - not only where it is displayed.

The Multi-Level Drill-Down Problem



As users drill from enterprise summary to individual performance analytics, the permission model must re-validate at every level - preventing a broad top-level grant from exposing restricted granular data below.



Data Freshness

Minute-Level Refresh, Zero Stale Permissions

Speed must never come at the cost of an incorrectly scoped permission state. Every data refresh cycle is paired with an authorization consistency check.

Keeping Live data and Access Control in Sync

Distributed Warehouse Architecture

Minute-level ingestion pipelines tied to access-aware query execution - scoped at read time, not batch time.

Permission-Stamped Data Tokens

Each data payload carries an authorization fingerprint, invalidated when the underlying policy changes - even mid-session.

Push Notification Gating

Proactive data pushes are permission-evaluated server-side before dispatch - no data reaches the device before authorization confirms eligibility.



AI-Augmented Insight Delivery

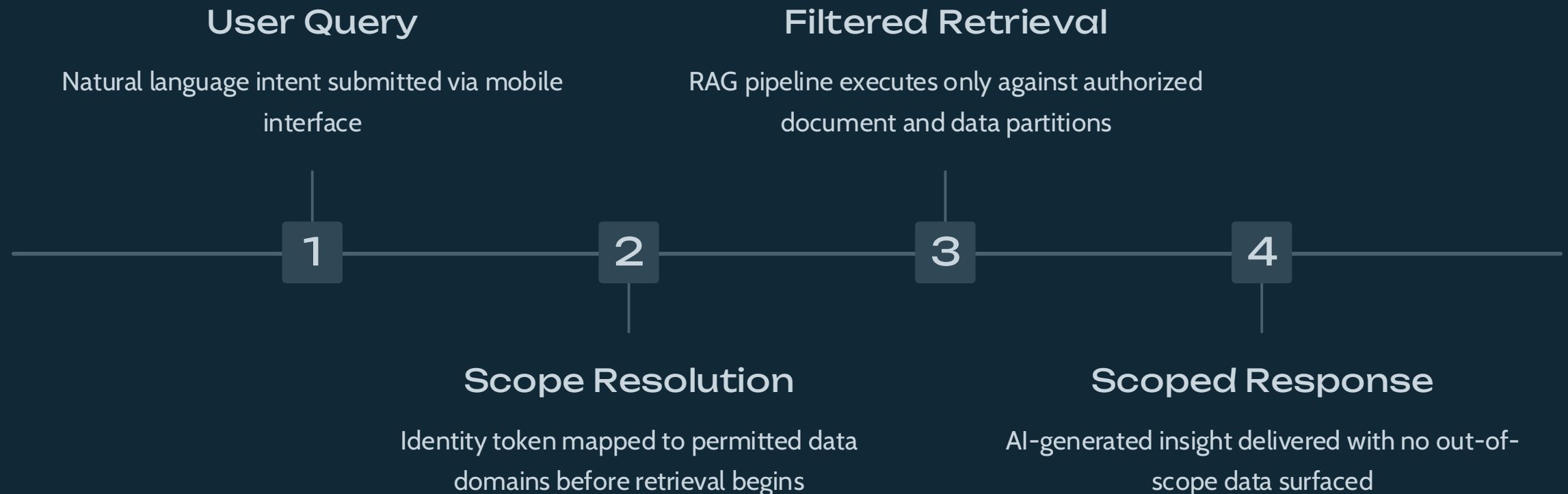


New Vectors, New Questions

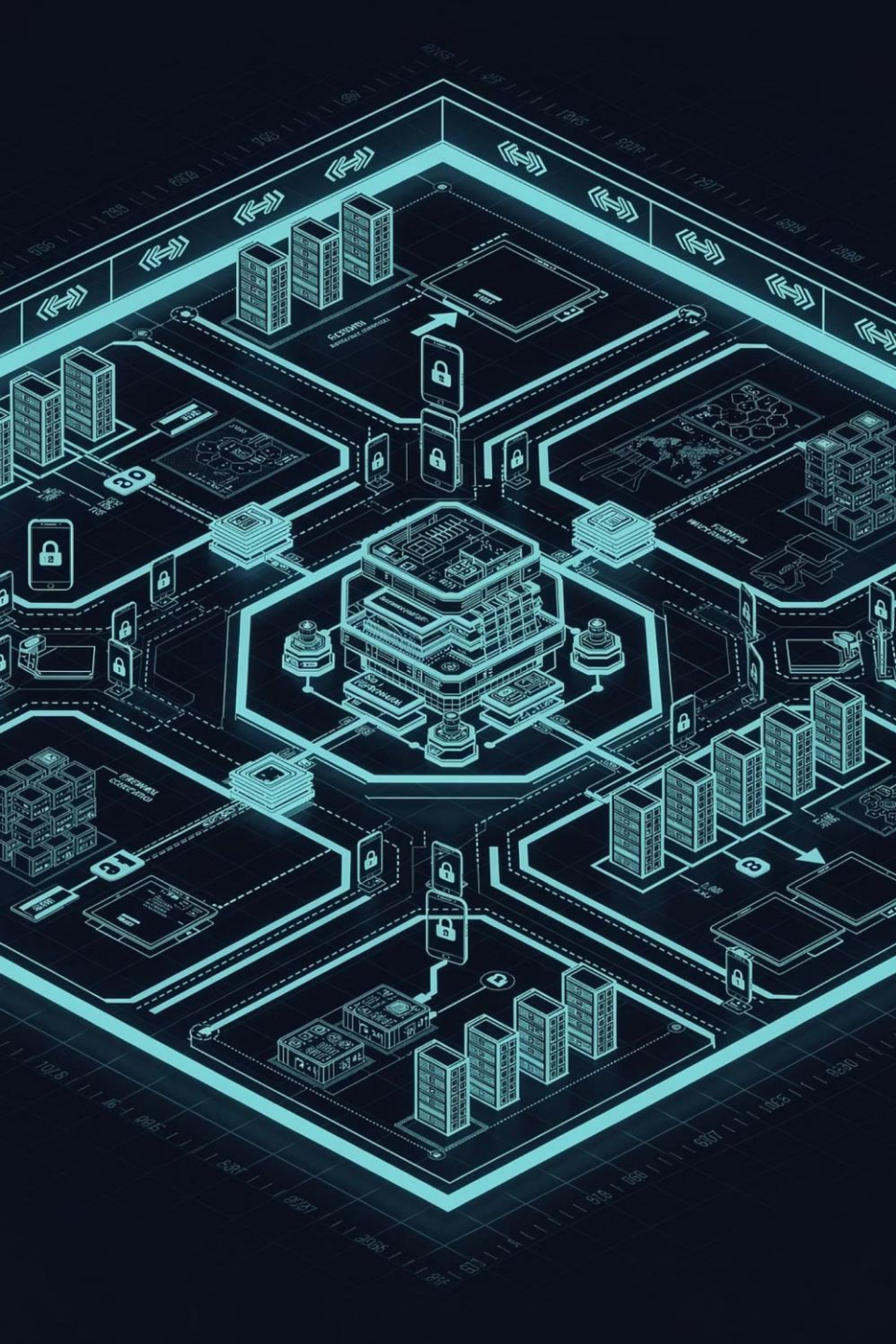
Push-based alerts and natural language querying extend how users reach operational data - but they introduce new authorization challenges that traditional RBAC models do not address natively.

- AI-generated summaries may aggregate data across scope boundaries
- Natural language queries are unbounded in intent - scope must be enforced at retrieval, not just display
- RAG pipelines must respect per-user data visibility at the vector search layer

Query-Scoped Authorization for Conversational Access



Query-scoped authorization must be treated as a first-class architectural concern - not a post-processing filter applied to AI output.



Blueprint

The Unified Architecture

Identity binding, access control, and low-latency performance are one design problem. The following principles anchor the reference architecture.

Key Takeaways: Your Blueprint

1 Bind identity at the device layer

Hardware attestation + federated SSO makes unmanaged devices trustworthy without MDM enrollment.

2 Cache permissions, not data

Edge-resolved, TTL-bounded permission caches deliver RBAC at mobile speed with fail-closed guarantees.

3 Enforce policy at integration, not presentation

Cross-domain data joins require federated policy - every drill-down level must re-validate scope.

4 Authorize the query, not just the user

AI and RAG-based delivery requires query-scoped authorization enforced at retrieval - not at output filtering.

Thank You